SLIDE 1:Scientific Instruments

The scientific output of the Aeris mission will primarily rely on data collected from two key instruments: a UV-Visible (UV-Vis) Spectrometer and the Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES). These instruments will work together to provide detailed insights into the interactions between Venus' atmosphere and its surface geology, focusing on the detection of carbonyl sulfide (OCS) gas and the identification of volcanic minerals such as pyroxenes. Data analysis will employ established remote sensing techniques tailored to the harsh environmental conditions of Venus, with interpretation based on comparative planetology and geophysical modeling.

The UV-Vis Spectrometer will produce high-resolution ultraviolet absorption spectra in the 185–300 nm range, targeting trace gases present in Venus' middle atmosphere. One gas of particular interest is carbonyl sulfide (OCS), a sulfur-bearing compound that serves as a possible indicator of active volcanism. OCS spectral features, notably those around 195–200 nm

The pHyTES instrument is designed to collect thermal infrared emission data within the 8–12 μm wavelength range to identify the mineral makeup of Venus' surface. This particular spectral window is well-suited for detecting silicate minerals such as pyroxenes, which are commonly associated with volcanic rocks.

SLIDE2:Data collection

Planned Sequence for Data Collection

After deployment from the spacecraft into Venus' atmosphere, the balloon inflates and starts a slow descent through various atmospheric layers. The following steps outline the experiment's timeline:

1.Initial stabilization (0–1 hour): The aerobot confirms proper orientation and altitude before collecting data.

2.Atmospheric gas scanning (1–12 hours): The UV-Vis spectrometer records gas absorption data in intervals, storing it locally.

3.Surface thermal imaging (3–20 hours): While passing over geologically interesting regions, the thermal spectrometer captures heat maps of the surface.

4.Data transmission: At regular intervals, information is relayed to an orbiter for delivery back to Earth.

5.Secondary pass (24–36 hours): Instruments re-scan to verify earlier results and detect changes over time.

6.Final phase (36–48 hours): As the balloon reaches lower, hotter layers, instruments shut down in sequence to preserve any last data.

Planning and Site Selection

This experiment plan supports a short but focused mission lifetime, maximizing scientific return while managing environmental risks. The systems work in harmony to achieve key measurements within the mission's 48-hour operational window.